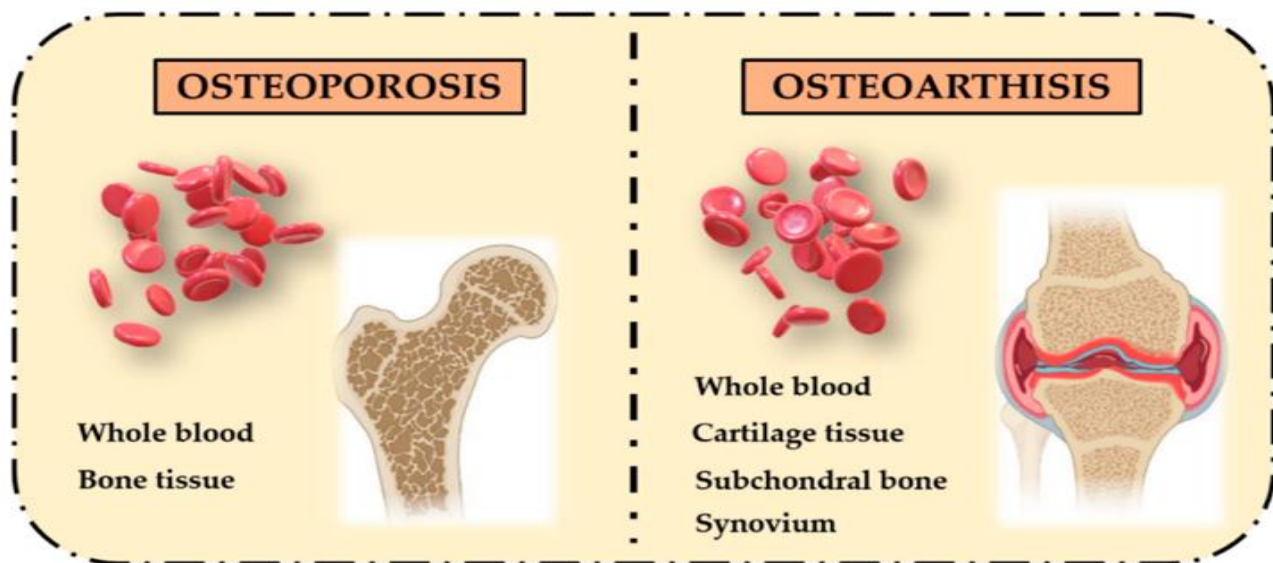


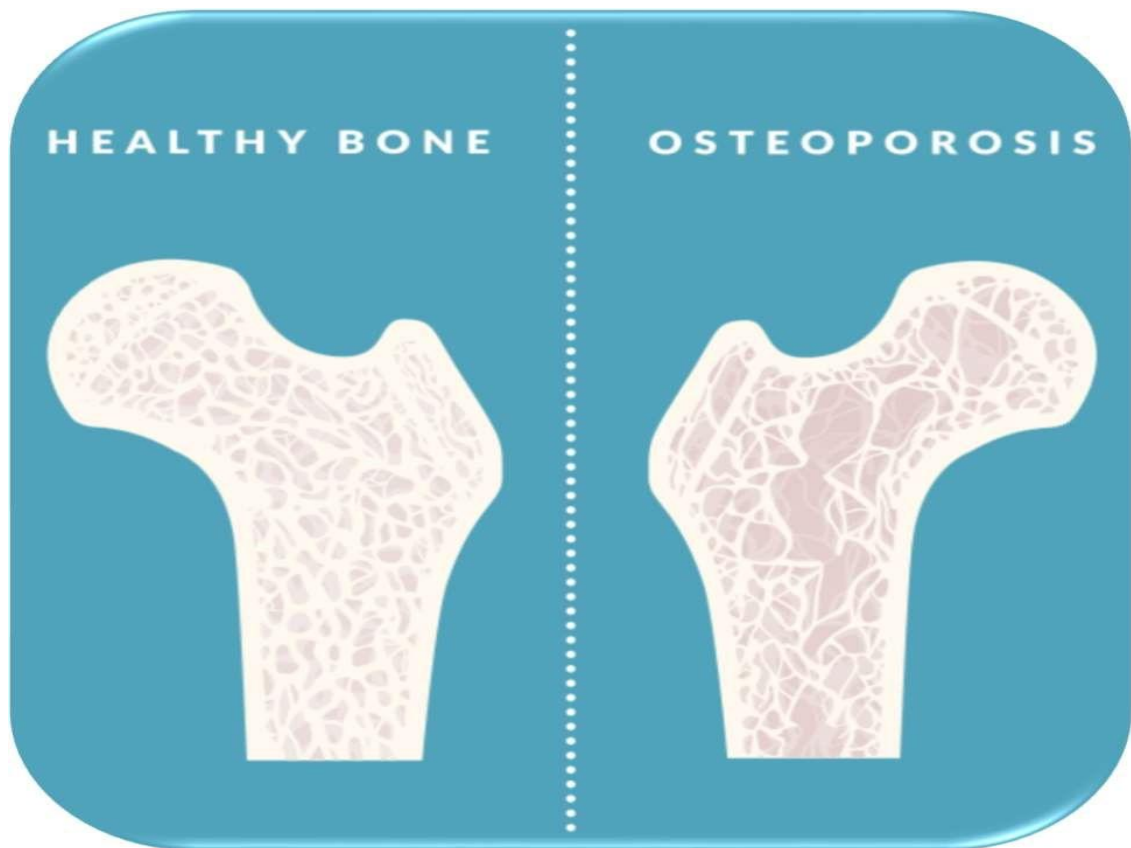
Osteoporosis & Osteoarthritis



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Osteoporosis



Introduction

Osteoporosis is a skeletal disorder characterized by low bone mass, damage of the bone tissue leading to bone fragility, and consequent increase in fracture risk. Osteoporosis becomes more common within age 50 and 80 are affected. It is more common in women than men.

Bones that commonly break include the vertebrae in the spine, the bones of the forearm, and the hip.

the person may have chronic pain and a decreased ability to carry out normal activities. It develops slowly over several years.

Definition

Osteoporosis is a bone disease occurs when the body loses too much bone tissue that cause bones becomes weak and may break from a fall or any high pressure on bones, making them fragile and more likely to break. osteoporosis also means “porous bone.”

Pathophysiology

Osteoporosis is a classic example of a multifactorial disease with a complex interplay of genetic, intrinsic, exogenous, and life style factors contributing to an individual's risk of the disease. Traditional pathophysiologic models frequently emphasized endocrine mechanisms, e.g. estrogen deficiency and secondary hyperparathyroidism in elderly due to estrogen deficiency, reduced dietary intake, and widely prevalent vitamin D deficiency, as the key determinants of postmenopausal osteoporosis.

Types

Primary :-

Type I osteoporosis (postmenopausal osteoporosis) generally develops after menopause, when estrogen levels drop precipitously. These changes lead to bone loss, usually in the trabecular (spongy) bone inside the hard cortical bone.

Type II osteoporosis (age - related osteoporosis) typically happens after age 70 and involves a thinning of both the trabecular (spongy) and cortical (hard) bone.

Secondary osteoporosis(disease or condition)

is very similar to primary except that it occurs in response to a particular disease, normally one that will affect hormone levels within the body such as conditions that interfere with thyroid health.

While primary may be addressed through a gradual change in activity levels and diet,

Osteogenesis Imperfecta : Osteogenesis imperfecta is a condition that is the result of a genetic mutation. The condition will normally exist with other symptoms such as respiratory issues,

Idiopathic Juvenile Osteoporosis: This pediatric condition has no known cause and usually has an onset just before puberty. In essence, it is brittle and porous bones with no other associated symptoms and will usually resolve without medical treatment after a relatively short amount of time.

If this condition is found to be present, it is recommended to have children monitor their activity or follow their physician's guidelines for maintaining general health.

Causes

Controllable causes:

- 1- Smoking. People who smoke lose bone density faster than nonsmokers.
- 2- Alcohol use. Heavy alcohol use can decrease bone formation, and it increases the risk of falling.
- 3- Getting little or no exercise.
- 4- Being small-framed or thin.
- 5- A diet low in foods containing calcium and vitamin D

Uncontrollable causes:

- 1- Being over age 50.
- 2- Being female.
- 3- Menopause.
- 4- Family history of osteoporosis.
- 5- Broken bones or height loss.

Risk factors

- Female gender, Caucasian or Asian race, thin and small body frames, and a family history of osteoporosis. (Having a mother with an osteoporotic hip fracture doubles your risk of hip fracture.)
- Cigarette smoking, excessive alcohol and caffeine consumption, lack of exercise, and a diet low in calcium.
- Poor nutrition and poor general health.
- Malabsorption (nutrients are not properly absorbed from the gastrointestinal system) from conditions such as Celiac Sprue.
- Getting little or no exercise.
- Being small-framed or thin.
- Chronic diseases such as rheumatoid arthritis and chronic hepatitis C, an infection of the liver.

- Immobility, such as after a stroke, or from any condition that interferes with walking.
- Hyperthyroidism, a condition wherein too much thyroid hormone is produced by the thyroid gland (as in Grave's disease) or is caused by taking too much thyroid hormone medication

Signs and Symptoms

Signs and symptoms of early-stage osteoporosis

Early, detectable signs of bone loss are rare. Often people don't know they have weak bones

until they've broken their hip, wrist, or some other bone.

However, some signs and symptoms can point toward potential bone loss, such as:

- Receding gums. Your gums can recede if your jaw is losing bone. Ask your dentist to screen for bone loss in the jaw.
- Weaker grip strength. researchers found that low handgrip strength was linked to low bone mineral density. In addition, lower grip strength can increase your risk for falls.
- Weak and brittle fingernails. Nail strength can signal bone health. But you should also take other factors into consideration that may affect your nails, such as exposure to very hot or cold temperatures, regular use of nail polish remover or acrylic nails, or submersion in water for long periods of time.
- Other than changes in bone density, osteoporosis doesn't usually cause a lot of initial symptoms. Your best bet for detecting it in the early stages is to talk with your doctor or healthcare professional, especially if you have a family history of osteoporosis.

Signs and symptoms of later-stage osteoporosis

Once bone mass has deteriorated further, you may start to experience more obvious symptoms, such as:

- Loss of height. Compression fractures in the spine can cause you to become shorter. This is one of the most noticeable symptoms of osteoporosis.
- Fracture from a fall. A fracture is one of the most common signs of fragile bones. Fractures can occur with a fall or from minor movement such as stepping off a curb. Some osteoporosis fractures can even be triggered by a strong sneeze or cough.
- Back or neck pain. Osteoporosis can cause compression fractures of the spine. These fractures can be very painful because the collapsed vertebrae may pinch the nerves that radiate out from the spinal cord. The pain symptoms can range from minor tenderness to debilitating pain.

Diagnosis

- Stooped posture or fracture. The compression of the vertebrae may also cause a slight curving of the upper back. A stooped back is known as kyphosis, which can cause back and neck pain. It can even affect breathing due to the extra pressure on the airway and limited expansion of your lungs.

Bone density scan (DEXA scan):

DEXA → Dual Energy X-ray Absorptiometry

used to measure your bone strength. It's a painless procedure that takes 10 to 20 minutes, and the amount of radiation you get from the X-rays the scan uses is low

it is considered about lie on an open X-ray table and try to stay still as the scanner passes over your body. and then you can go home.

The bone density difference is calculated as a standard deviation (SD) and is called a T score. A T score of:

- above -1 is normal.
- between -1 and -2.5 shows bone loss and is defined as osteopenia
- below -2.5 shows bone loss and is defined as osteoporosis.

Treatment

There isn't a cure for osteoporosis. However, treatment is available to help slow the progression of the disease and manage symptoms.

- **Non pharmacological treatment:**

- Eat habits: bones need calcium and vitamin D to stay strong and healthy. Not getting enough calcium early on in life can lead to osteoporosis later on.

- Life style: treatments for established osteoporosis may include exercise. Exercises are often suggested to help you prevent osteoporosis. Weight-bearing, resistance and balance exercises are all important.

- Decrease the smoking and alcohol.

- **Pharmacological treatment:**

- 1.vitamin D supplement

- 2.Antiresorptive medications slow the rate of bone loss.

- 3.Anabolic medications promote bone growth.

Complications

1. increase susceptibility for break and fracture

2. Limited mobility: physical activity. A loss of activity can cause you to gain weight.

3. Depression: Less physical activity can lead to a loss of independence and isolation. Activities you once enjoyed may be too painful now. This loss, added to the possible fear of fractures, can bring on depression.

4. pain: Fractures caused by osteoporosis can be severely painful and debilitating

Prevention

Primary:

Health education :-

1. increase in calcium intake
2. increasing appropriate physical activity
3. smoking cessation
4. exposure to sunlight,
5. weight management,
6. alcoholic beverage ¶drinking control
7. prevention from falling

Health promotion:

Diet: High protein ,calcium intake ,vitamins &minerals Life style :Exercise, waking half hour every day.

Specific protection:

Increase exercise & healthy diet calcium supplement & vitamin D

Secondary:

Early diagnosis: Low energy fractures, Stooping and height loss, Bone pain ,Joint pain can be diagnosed by Bone density scan (DEXA scan): and proper treatment: estrogens/gestates.

Selective estrogen receptor modulators in postmenopausal women.

Tertiary:

Follow up: every 3m

Prevent complications: reduce the femur fractures.

In addition to drugs such as vitamin D/calcium, vitamin D metabolites and bisphosphonates.

Rehabilitation: refer to occupational to adopt with problems.

Nursing management

1- Diagnosis : Acute Pain

Related to :

- Fractures.
- Muscle spasm

Interventions:

Allow patient to maintain a diary of pain ratings, timing, precipitating events, medications.

Recognize and convey acceptance of the patient's pain experience.

Aid the patient in making decisions about choosing a particular pain management strategy.

Administer medications and treatments to improve these functions.

Allow the patient to describe appetite, bowel elimination, and ability to rest and sleep.

Discuss patient's fears of undertreated pain, addiction, and overdose.

2-Diagnosis : Imbalanced Nutrition: Less Than Body Requirements.

Interventions:

- Set appropriate short-term and long-term goals.
- Provide a pleasant environment.
- Promote proper positioning.
- Provide good oral hygiene and dentition.
- Provide companionship during mealtime.
- Consider six small nutrient-dense meals instead of three larger meals daily to lessen the feeling of fullness.
- For patients with physical impairments, refer to an occupational therapist for adaptive devices.
- Determine time of day when the patient's appetite is at peak. Offer highest calorie meal at that time.
- Offer high protein supplements based on individual needs and capabilities.
- Offer liquid energy supplements.
- Keep a high index of suspicion of malnutrition as a causative factor in infection.